To operationalize this in your library, we will structure each step of the Weighted-State workflow as a **Machine-Readable Procedure (MRP)**. Each MRP is designed to be a "fillable-form" module that your SIS orchestrator calls sequentially.

1. The MRP Library: Parametrized Workflow

Step 1: Integrity Poll (MRP_001)

This procedure performs the health check across the mesh.

```
{
  "ProcedureID": "MRP_001_IntegrityPoll",
  "Inputs": ["Link_Registry_List"],
  "Logic": "Query each Ghost Pointer for Eigenstate stability.",
  "Output": "Integrity_Status_Map",
  "Fields": {
    "Link_ID": "UUID",
    "Correlation_Value": "Float_0_to_1",
    "Stability_Score": "Timestamped_Variance"
  }
}
```

Step 2: Priority Map (MRP_002)

Maps the incoming data payload requirement to the necessary Mesh Service Level Agreement (SLA).

```
{
  "ProcedureID": "MRP_002_PriorityMapping",
  "Inputs": ["Payload_Type", "Integrity_Status_Map"],
  "Logic": "Assign required Link-Weights based on Payload_Type.",
  "Output": "Active_Routing_Table",
  "Fields": {
    "Required_Integrity_Threshold": "Float",
    "Min_Link_Count": "Integer",
    "Redundancy_Profile": "String_Enum"
  }
}
```

Step 3: Link Selection (MRP_003)

The gatekeeper that selects the physical endpoints for the current transmission.

```
{
  "ProcedureID": "MRP_003_LinkSelection",
  "Inputs": ["Active_Routing_Table", "Integrity_Status_Map"],
  "Logic": "Select top-N links exceeding the threshold defined in
MRP_002.",
  "Output": "Selected_Path_List",
```

```
  "Fields": {
    "Selected_Links": ["Link_ID_Array"],
    "Weight_Allocation": "Map<Link_ID, Priority_Score>"
  }
}
```

Step 4: Sync Dispatch (MRP_004)

The atomic write operation that bridges the state to the selected mesh.

```
{
  "ProcedureID": "MRP_004_SyncDispatch",
  "Inputs": ["Payload_Data", "Selected_Path_List"],
  "Logic": "Execute parallel state-write across selected links.",
  "Output": "Dispatch_Receipt",
  "Fields": {
    "Timestamp": "ISO_8601",
    "Transaction_Signature": "Hash_Of_State_Change"
  }
}
```

Step 5: Post-Verification (MRP_005)

The consensus validation that ensures parity across the hub.

```
{
  "ProcedureID": "MRP_005_Verification",
  "Inputs": ["Dispatch_Receipt", "Selected_Path_List"],
  "Logic": "Compare local write against remote read parity.",
  "Output": "Final_Integrity_Result",
  "Fields": {
    "Parity_Check": "Boolean",
    "Mismatch_Details": "Exception_Trace"
  }
}
```

The Orchestrator "Form"

In your AARM architecture, you can now generate a single **Unified Workflow Form** that calls these five procedures as a batch.

Usage Instruction for the Library:

1. **Instantiate:** Load the Workflow_ID into the SIS registry.
2. **Fill:** Insert the Payload_Data and Target_Node_Coordinates into the header.
3. **Execute:** The system iterates through the five MRPs, maintaining local buffering at each step if the Parity_Check (Step 5) fails, triggering the CAR_Handler (from our previous discussion) to re-initialize the specific link that failed.

This modular structure allows you to swap out individual procedures (e.g., changing the

Consensus_Protocol in MRP_005) without rewriting the entire pipeline, keeping your codebase clean, generic, and highly maintainable.

Would you like to define the **Exception Handling** logic for when MRP_005 (Verification) fails, specifically regarding how the system should handle partial state updates?